



Letter to the Editor

Response to “Rethinking ECT in Depression Among the Oldest-Old: An Underexplored Opportunity”

Dear Editor,

We thank Drs Flôr-Rodrigues, Howell Lole Harris, and Koizia for their thoughtful letter regarding our article,¹ and their kind remarks about the timeliness and potential importance of this work.² We appreciate the opportunity to further discuss the vulnerable and growing group of oldest-old patients with depression.

We agree with their comment on the limitations of register-based research. Such real-world evidence has several strengths, including large, population-representative samples, treatment results from routine clinical practice, follow-up spanning several years before and after treatment, and the opportunity to take a holistic perspective on demographics and healthcare use received by patients.^{3,4} At the same time, there are important limitations that must be addressed through complementary research to strengthen the overall evidence base. We agree and strongly encourage further studies with more in-depth

assessments of potential adverse physiological or cognitive effects, including cardiovascular abnormalities and memory dysfunction. Such work would expand knowledge and increase confidence in recommending ECT as a valid treatment for depression in the oldest-old.

As Dr Flôr-Rodrigues with colleagues note, a further limitation of register-based research is the lack of comprehensive measurement of potential confounders. We used advanced statistical methods to mitigate indication bias, but agree that some residual confounding is likely to persist. However, given the widespread use of ECT in Sweden and its well-established safety profile, including in frail elderly patients, we believe that the extent of indication bias in our data is likely smaller than in many other healthcare settings.

Although potentially challenging in practice given the small target population, we believe a randomized controlled trial of ECT in depressed oldest-old patients would be valuable, as this design addresses many limitations inherent to observational research.⁵ However, among the most severely ill patients, those for whom ECT tends to be most effective, the requirement to provide informed consent to randomization is particularly challenging. Delaying an effective

treatment in the presence of suicide risk or increased mortality raises ethical concerns, and cognitive impairment as well as ambivalence may further complicate the consent process. Consequently, observational studies are needed to complement randomized trials in ECT research.

We also appreciate the comment regarding education. ECT is already a part of all undergraduate medical training in Sweden, and to independently perform ECT is in the core curriculum to become a psychiatrist. However, we agree that further integration of ECT teaching in medical curricula would support better-informed clinical decision making, and particularly for subpopulations where uncertainty and caution may be greatest, such as the oldest-old. In addition, we support the ambition to extend education efforts to the general public. While of limited international reach, our findings were covered positively in Swedish local news outlets, including a magazine for retirees. We believe that increasing knowledge among the general population of oldest-old may also help reduce barriers to accessing ECT among depressed oldest-old patients.

In summary, we are grateful for this collegial engagement and agree that further research and education on the safety and efficacy of ECT in oldest-old patients

with depression are needed. We hope this exchange helps stimulate continued research on this patient group.

DECLARATION OF GENERATIVE AI AND AI- ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

ChatGPT (OpenAI) was used for proofreading. No content or conclusions were generated by the tool. The authors reviewed and approved all changes and take full responsibility for the content.

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